Lecture 3

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Soil Mechanics

- Soil mechanics is a branch of soil physics and applied mechanics
- **Karl Terzaghi** is popularly known as the father of Soil Mechanics.
- Soil mechanics is defined as the study of the mechanical behavior of soil, particularly concerning its structure, stability, and resistance to deformation under stress.
- It encompasses the understanding of soil pore structure and the impacts of mechanical forces such as compaction and erosion.

Application

Foundation design

Ensuring buildings, bridges, towers, and roads rest safely on soil.

Earthwork and Embankment Construction

Dams, levees, road embankments, railways, landfills.

Slope Stability Analysis

Preventing landslides and designing stable cut/fill slopes in highways and hill areas.

Pavement Design

Determining subgrade strength for flexible and rigid pavements.

Application

Retaining Structures

Designing retaining walls, sheet piles, and earth-retaining systems.

Groundwater Flow & Drainage

Analyzing seepage, dewatering systems, and drainage design for construction sites.

Soil Improvement Techniques

Stabilization, compaction, geosynthetics, grouting, ground reinforcement.

Environmental and Geotechnical Projects

Landfills, contaminated site remediation, erosion control.

❖ *Shear Failure-Loads have exceeded shear strength capacity of soil!*



Transcosna Grain Elevator, Canada Oct. 18, 1913





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Physical Properties of soil

Soil physical properties affect the appearance and feel of a soil.

The major soil physical properties are:

- Soil Texture
- Soil Structure
- Soil Consistence/Soil Strength
- Soil Color
- Soil Permeability
- Soil Temperature

Soil texture:

Each soil separate represents a distinct physical size group. Mineral particles less than 2 millimeters in equivalent diameter and ranging between specified size limits.

Soil texture refers to the relative proportion of three different-sized mineral particles in soil: sand, silt, and clay.

The names and sizes of the soil separates recognized in the United States are as follows

General Classification

Sand	2.0 - 0.05 mm
Silt	0.05 - 0.002 mm
Clay	<0.002 mm

Very Coarse Sand	2.0 - 1.0 mm
Coarse Sand	1.0 - 0.5 mm
Medium Sand	0.5 - 0.25 mm
Fine Sand	0.25 - 0.10 mm
Very Fine Sand	0.10 - 0.05 mm
Silt	0.05 - 0.002 mm
Clay	0.002 mm

Sand

2 to 0.05 mm



Particles visible without a magnifying glass

Feels coarse and gritty

Silt

0.05 to 0.002 mm



Not typically visible by the unaided eye

Feels smooth and floury

Clay

< 0.002 mm



Seen only with an electron microscope

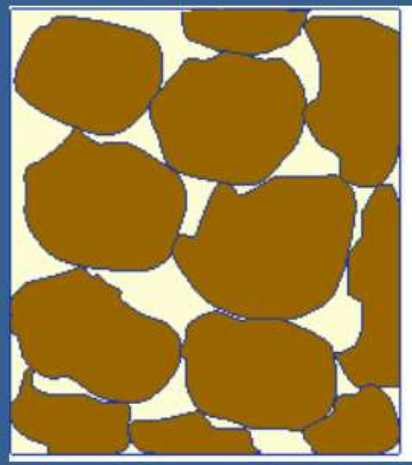
Feels stiff and sticky

Soil Texture - Particle-Size Distribution



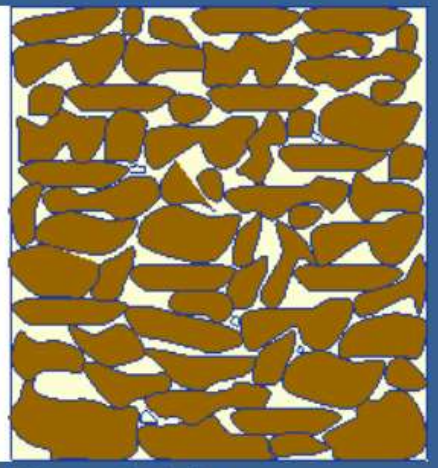
Soil Mineral Particles Settled in Cylinder

Coarse Textured Soil

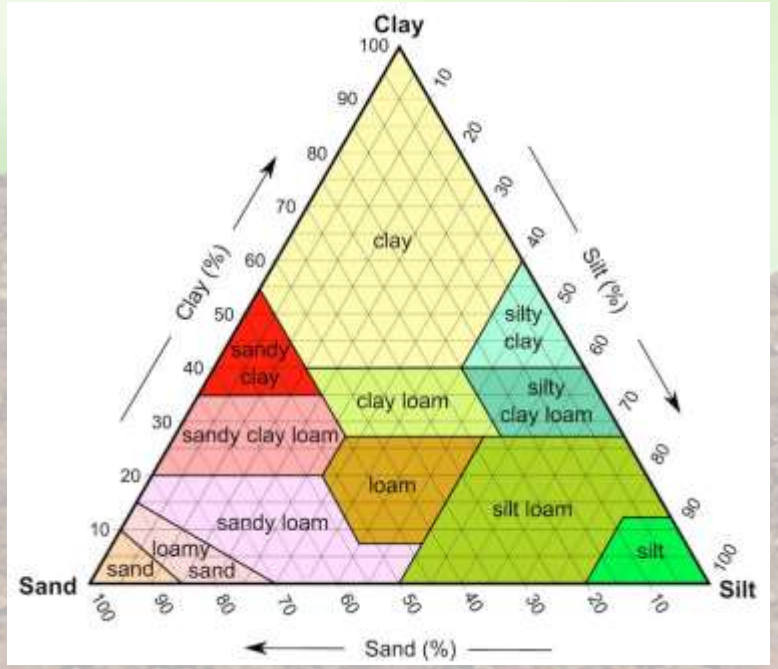


Less porespace but more macropores

Fine Textured Soil



More total porespace



Soil Structure

- Structure is the arrangement of primary sand, silt and clay particles into secondary aggregates called peds or structural units which have distinct shapes and are easy to recognize. These differently shaped aggregates are called the structural type.
- Soil structure and texture affect how the soil will hold water and nutrients.
- The five basic types of structural units are as follows:

a. Platy:

- Plate-like aggregates that form parallel to the horizons like pages in a book.
- This type of structure **may reduce air, water and root movement**.
- common structure in an E horizon and usually not seen in other horizons.

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Soil Structure

b. Blocky:

- Two types-angular blocky and subangular blocky
- These types of structures are commonly seen in the **B horizon.**
- Angular is cube-like with sharp corners while subangular blocky has rounded corners.

c. Prismatic:

Vertical axis is longer than the horizontal axis. If the top is flat, it is referred to as **prismatic.**

If the top is rounded, it is called **columnar.**

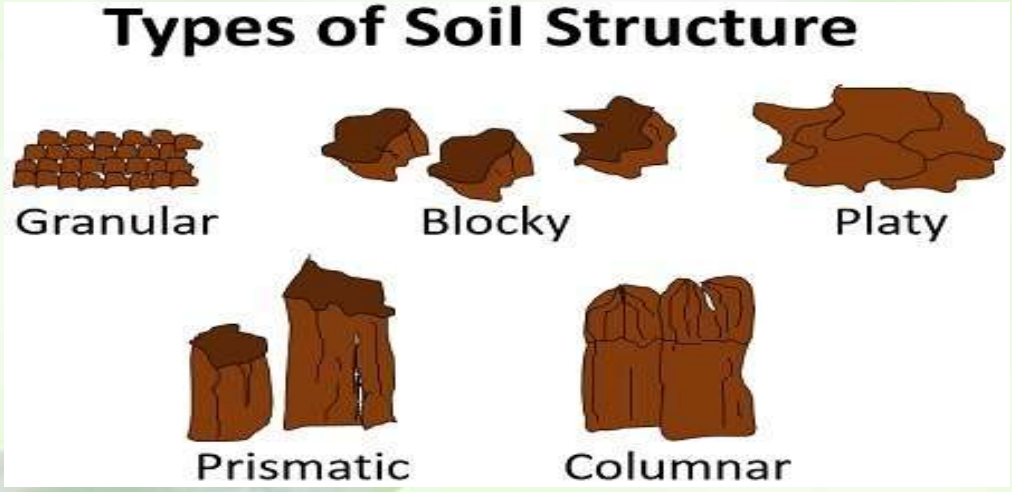
d. Granular:

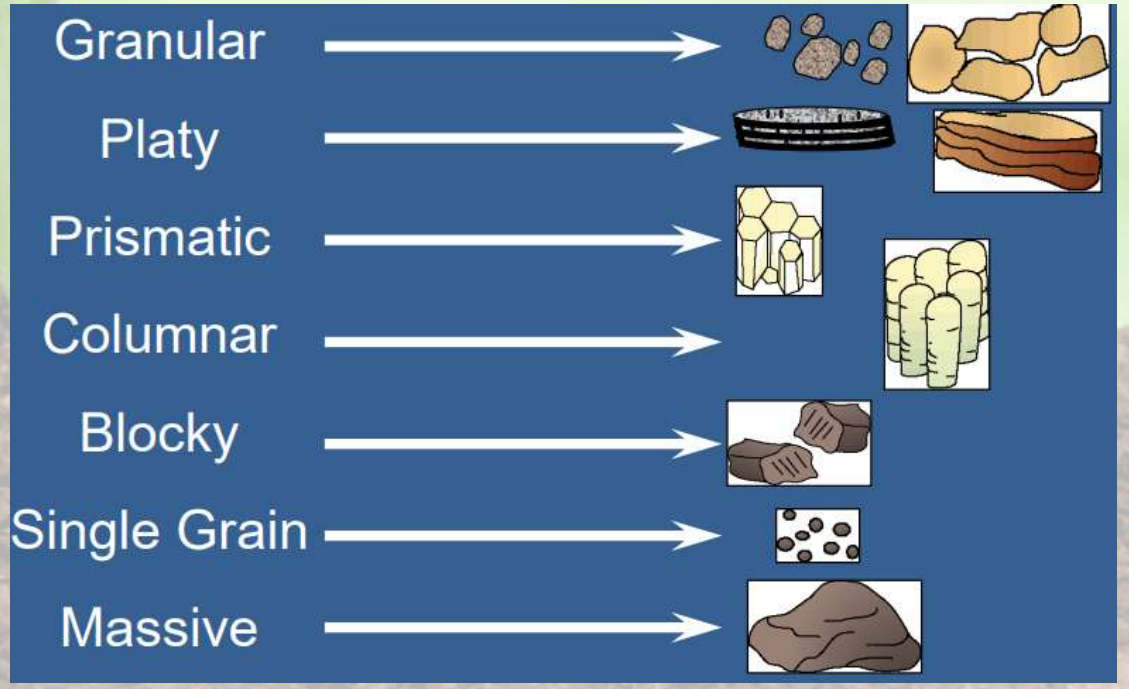
Peds are round and pourous, spheroidal. This is usually the structure of A horizons.

Soil Structure

e. Structureless

- No observable aggregation or structural units.
- Single grain-sand
- Massive-solid mass without aggregates





Soil colour:

It is the most obvious and easily determined soil property It has little direct effect

on the soil, but is an indicator of soil properties. However, there are many things we can

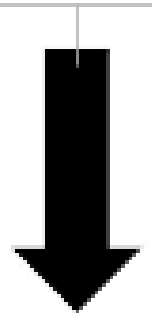
tell about the soil by observing the color.

- Organic matter content; the more organic content the darker the soil color
- Soil color and soil temperature : dark colored soils absorb more heat so they warm up quicker and have higher soil temperatures.
- Soil color and parent material : generally dark parent material will develop into dark soils.

Soil Permeability

- Permeability is the speed of air and water movement in a soil - this is affected by texture and structure

1. if permeability is high : water moves quickly
2. if permeability is low : water moves slowly

Soil type	Permeability k (m/sec)	Reducing permeability
Gravel	>0.1	
Sand	$0.1 - 10^{-4}$	
Silt	$10^{-4} - 10^{-8}$	
Clay	$<10^{-8}$	

Soil Consistency

- **Soil consistency** is the strength with which soil materials are held together or the resistance of soils to deformation and rupture.
- Consistency is a term which is used to describe the degree of firmness, cohesion, and resistance of soil.
- **Consistency of a soil changes with the amount of water.**
- **Soil consistency is determined for wet, moist and dry soil samples. In case of wet soils, it is expressed as both stickiness and plasticity. Soil consistency may be estimated in the field using simple tests or may be measured more accurately in the laboratory.**
- **a. Wet:** Stickiness (non-sticky, slightly sticky, sticky, very sticky)
- Plasticity (non-plastic, slightly plastic, plastic, very plastic)
- **b. Moist:** Very friable, friable, firm ,very firm
- **c. Dry:** Loose, soft. slightly hard, hard, very hard ,extremely hard
- **It indicates amount and type of clay material or fine grain soil**

Four state of soil consistency

- A swedish agriculture engineer **Albert Atterberg** mentioned that depending upon the water content the consistency of fine grained soils can be described in four states.
- Liquid State
- Plastic State
- Semi-solid State
- Solid State
- In each state, the consistency and behavior of a soil is different and consequently so are its engineering properties. Thus the boundary between each state can be defined based on a change in the soil's behavior.

Consistency limit and Atterberg limit

- The water content at which the soil changes from one state to other is known as the **Consistency limit**.
- Atterberg demonstrated the significance of these limits in understanding the behaviour of clays. So these limits are also called **Atterberg limits**.
- **Atterberg Limits** are a set of moisture contents that define the boundaries between the different consistency states of fine-grained soil.
- Atterberg's limits are used to measure the physical condition of soil at different water contents. These depend on texture, organic matter content and amount of clay in the soil.

Soil consistency states

- **Liquit state**: When we mix fine grained soil in large quantities of water the resulting suspension of soil is called as liquid state of soil. In this state soil has virtually **no shear strength** which means soil has zero shear strength. Also it offers practically **no resistance** to flow and It behaves like a thick liquid or slurry.
- If the water content of the suspension is reduced, the soil becomes stiffer and starts developing resistance to shear.
- This is the stage when the sample changes from possessing no shear strength to possessing an infinitesimal shear strength and changes from the liquid to plastic state.

Liquid limit and plastic limit

The water content at which soil changes from the liquid to plastic state is known as **liquid limit**.

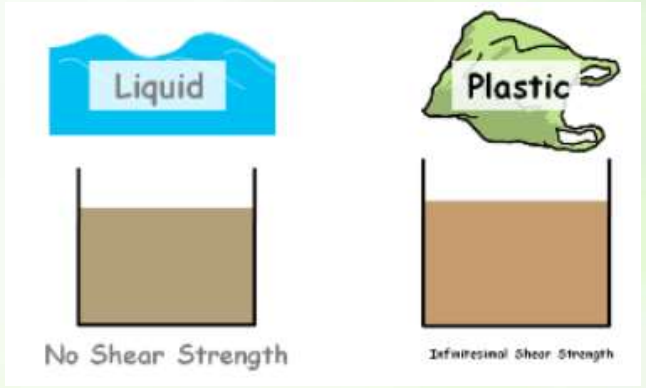
In other words the water content at which soil stops being liquid is its liquid limit. It does not flow like a liquid anymore.

In plastic state soil can be moulded into different shapes without rupturing it due to its plasticity.

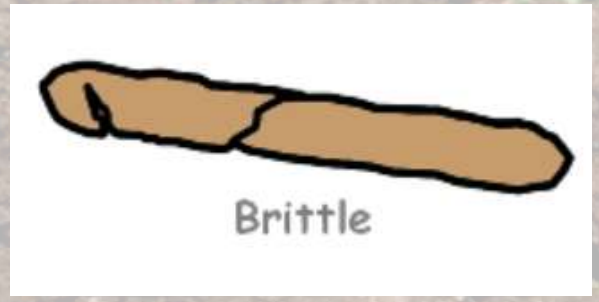
Now if we further reduce the water content of the soil, its plasticity decreases and finally soil changes its state from plastic to semi-solid.

In this state if we try to mould the soil, it cracks. Soil loses its plasticity and becomes brittle.

The water content at which the soil stops being plastic and changes to semi-solid state is called **plastic limit** of the soil.



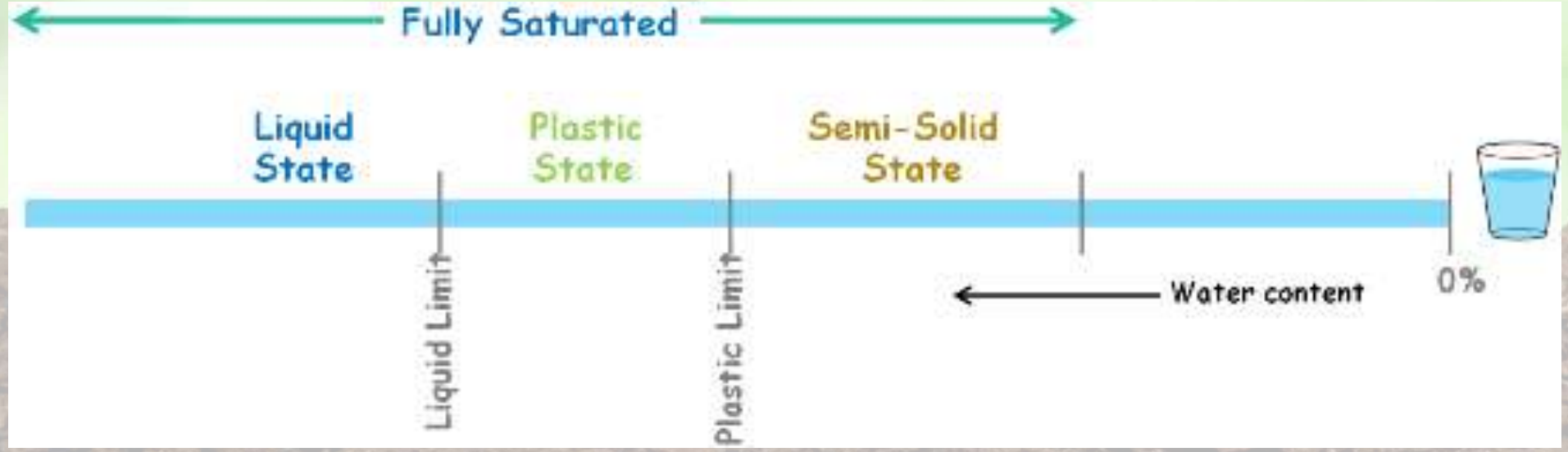


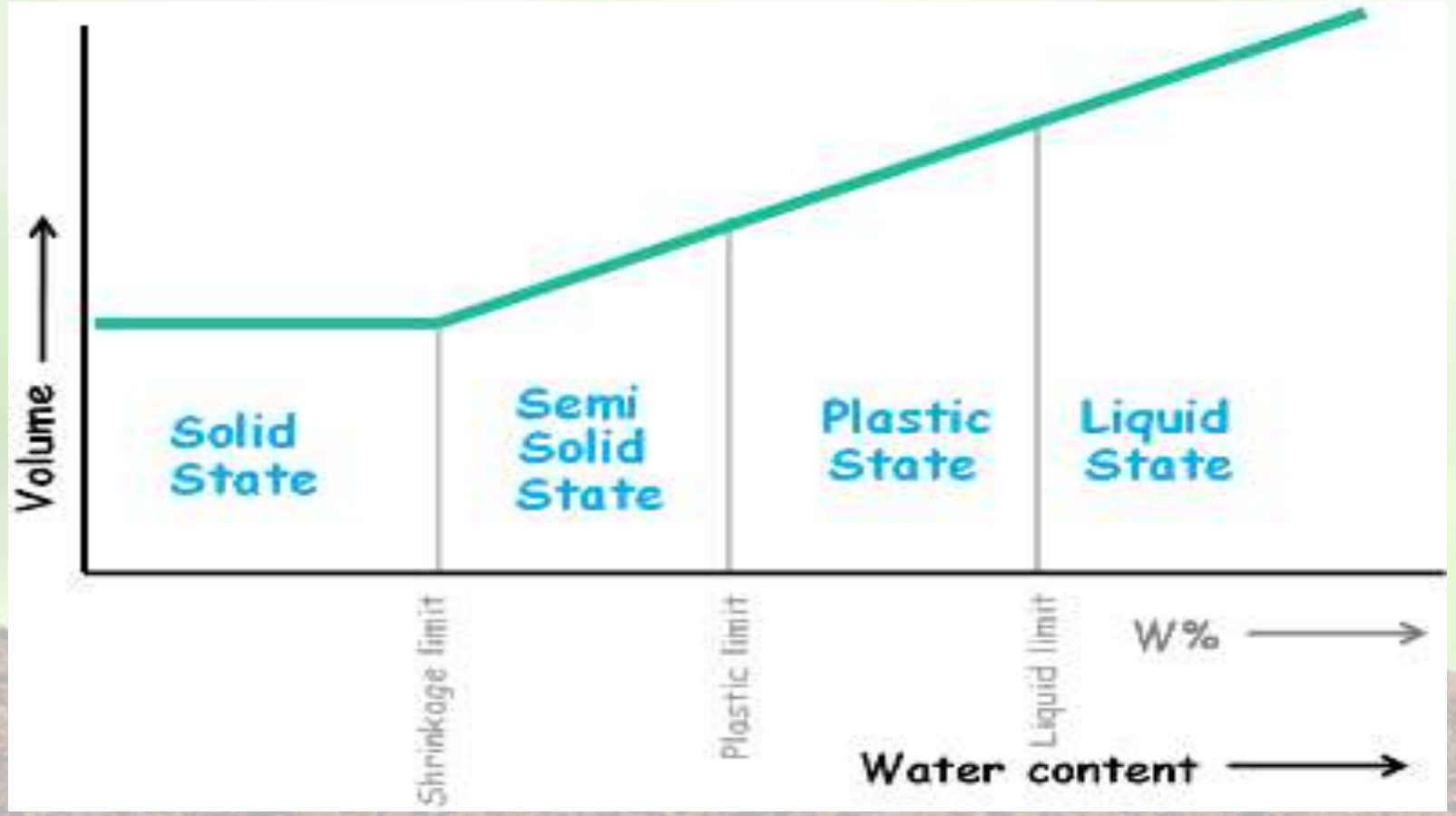


Shrinkage limit

- Up to the semi-solid state, the soil stays fully saturated, and its volume reduces nearly in the same proportion as water is lost.
- But there comes a stage when further reduction in water content, volume of soil remains the same. At that point soil mass changes from semi solid state to solid state.
- Soil converts into a soil mass which has pores partially filled with water. Now we can understand that if we reduce the water content, soil's volume remains the same but it's pore water gets reduced. Hence no volume change with water content reduction. That is soil does not shrink any more.
- And the water content at which soil stops shrinking, is called its **shrinkage limit**.
- Shrinkage limit can also be defined as the lowest water content at which soil is fully saturated.

- If we further reduce the water content below this limit sample begins to dry up at the surface and soil is no longer fully saturated. The colour of the sample also begins to change.





Explain the graph